Proof Point AI

Week 3 Check-in: Users & Data

Project detail	Information
Track	Research Paper Hub
Principle	**Transparency**
Product	Chatbot that help 3rd and 4th students to begin the research journey by helping them to search for relative papers and review their papers and make sure it out of issues.
User group	First-time thesis 3rd and 4th students at Arabic-speaking universities who are writing their first literature review in English (as proposed for approval; update if the instructor changed it).
Team name	**ProofPoint AI**
Team members	1. Ahmed Hazem Abdelalim Elabady 2. Mazen Hisham Mostafa 3. Alaa Osama Mohamed 4. Omnia Mohamed Ahmed
Repository	MazenHisham1/Research-Paper-Discovery-Hub
Check-in date	23 Oct 2026

1. Three fictional personas

Coverage check: the three personas span below-typical, typical, and advanced technical levels; two are novice AI users; two have an accessibility need; and they come from different cultural backgrounds and fields. The final personas are the team's own synthesis, edited from AI drafts (see Appendix A).

Requirement	Nour	Oliver	Hana
Technical level	Below typical	Typical	Advanced
AI experience	Novice	Novice	Experienced
Accessibility needs	Cognitive-load sensitivity	None noted	Low vision
Background	Egyptian, Arabic-speaking	Nigerian, English-speaking	Japanese, bilingual
Relation to approved user group	First-time thesis student	Master's student writing a thesis-proposal literature review	Research assistant

Persona 1: Nour Hassan

Field	Description
Age and role	22, final-year pharmacy student writing her graduation project.
Background	Grew up in an Egyptian city; first in her family to write a research report in English. Arabic is her first language; she reads English slowly.
AI experience	Novice. She has never used a chatbot for studying and is unsure what it can do.
Accessibility needs	Cognitive-load sensitivity. Needs plain language, short sections, clear headings, one step at a time, and control over how much information is shown, with concise explanations of unfamiliar research terms.
Device and connection	Shares a phone with her sister; mobile data is limited, so she avoids heavy pages.
Goals	Find five good papers for her project and understand them well enough to explain them to her supervisor.
Frustrations	English abstracts are dense. She does not know which papers matter. She worries about copying the wrong source.
Feelings about AI	Curious but cautious. She fears being blamed if the AI is wrong.
How Proof Point helps	Disclosure at the start, plain-language summaries, a "Show evidence" button, and an Arabic interface option.

2: Oliver Noah

Field	Description
Age and role	21, third-year civil engineering student preparing to write his first literature review for a university research project.
Background	From Nigeria, studying abroad. Comfortable in English; new to the local university system and its library tools.
AI experience	Novice with AI for research. He tried a general chatbot once and received a reference that did not exist.
Accessibility need	None noted.
Device and connection	Laptop; university Wi-Fi that drops during peak hours.
Goals	Build a reliable reading list quickly before a proposal deadline.
Frustrations	Cannot tell real citations from invented ones. Wastes time checking each reference.
Feelings about AI	Skeptical. He will trust it only if he can verify the source himself.
How Proof Point helps	Every claim links to the exact passage, and the bot says "I am not sure" when evidence is weak. Saved list with sources.

Persona 3: Hana Sato

Field	Description
Age and role	22, fourth-year data science student preparing her first literature review for a graduation research project.
Background	From Japan; fluent in English and learning Arabic. Uses several scholarly search tools daily.
AI experience	Experienced. Uses AI tools weekly for coding and reading.
Accessibility need	Low vision. Uses a screen reader and 200% zoom. Needs headings, landmarks, announced new messages, keyboard use, and good contrast.
Device and connection	Laptop with a screen reader; stable connection.
Goals	Scan many papers fast and verify AI claims before using them.
Frustrations	Interfaces that break with zoom or do not announce new content. Tools that hide how a result was chosen.
Feelings about AI	Positive but demanding. She expects proof, not confidence.
How Proof Point helps	Evidence panel with retrieval score, full keyboard control, live-region announcements, and editable search terms.

2. Bias-check notes

What the team reviewed: names, jobs, examples, images, contexts, language, and how each persona is described. We used a checklist and compared the first AI drafts with the final versions. Log entry numbers refer to Appendix A.

What we reviewed	Stereotype or exclusion found	What we changed
Names	First AI drafts used only common names from one region and paired them with matching jobs.	Chose names from different regions and checked that names do not decide a persona's job or ability.
Jobs and settings	The first Claude draft (entry 1) wrote the Arabic-speaking student as struggling because of her culture.	Reframed the difficulty as reading in a second language and limited data, not culture.
Disability	The first ChatGPT draft (entry 2) described the low-vision persona as needing help and pity.	Made her an advanced, capable user with specific interface needs.
Gender	Women were placed in caring or support roles in the first drafts.	Placed the personas in pharmacy, engineering, and data science with varied genders.
Connection and device	Assumed poor internet for students from smaller cities.	Tied connection limits to a specific situation (shared phone, Wi-Fi drops), not to a place.
Examples and images	Sample topics all came from Western health and tech contexts.	Added topics from engineering, pharmacy, and data science; used no photos of real people.
Language and religion	Greeting examples assumed one cultural style.	Used neutral greetings and made tone settings user-adjustable.

Done: Arabic summaries were checked with a native speaker, and names and phrasing swaps were tested in the fairness test set.

3. Data source description

Item	Description
What it is	OpenAlex: an open catalog of scholarly works (titles, authors, years, citation links, abstracts), accessed through its public API.
How we use it	We use it to search for needed papers by titles, abstracts, and metadata for topics like (machine learning in healthcare, renewable energy storage, water treatment, education technology, public health) and index them for search.
Size	We didn't have any specific size number since we use it as dynamic tool that its input changed by changing the needed paper type.
License and permission	**Verified by the team on 26 October 2026** against the official OpenAlex documentation: developers.openalex.org (overview page) and help.openalex.org/data/licenses. Findings: the dataset is released under the CC0 license (no rights reserved). The API is free but needs a free API key, with about $1/day of free usage; the downloadable data snapshot is free and updated quarterly. OpenAlex asks that research using it cite Priem, Piwowar and Orr (2022), arXiv:2205.01833. We will cite it in our report. Licensing of any abstracts or full text taken from other sources is checked separately.
Personal data	Scholarly publication metadata only. Author names are kept only to identify and cite papers. No personal user information is collected.
Why it fits the track	Paper identifiers, links, and abstracts let users trace each AI-generated answer back to its source and check that summaries match the original research.
Limits	Mostly abstracts, not full text; some records have no abstract. Language and geographic coverage will be examined, not assumed representative. Limits will be shown to users, and the chatbot will not present unsupported conclusions as established facts.

4. Prompt log (Appendix A)

#	Tool	Exact prompt	What it gave	What we changed and why
1	Claude	Create three fictional personas for a chatbot that helps first-time thesis students find research papers. Different technical levels (below typical, typical, advanced). At least two are novice AI users and one has an accessibility need. Varied cultural backgrounds. Include goals, frustrations, device and connection, and feelings about AI. Avoid stereotypes.	Three personas in a table.	Rewrote names and settings, removed culture-based explanations of difficulty, and added specific constraints (shared phone, Wi-Fi drops).
2	ChatGPT	Create three fictional personas for a chatbot that helps first-time thesis students find research papers. Different technical levels (below typical, typical, advanced). At least two are novice AI users and one has an accessibility need. Include varied cultural backgrounds. Include goals, frustrations, device and connection, and feelings about AI. Avoid stereotypes.	Three personas with more detail on daily routines.	Compared against the assignment requirements. Removed descriptions that portrayed the person with a disability as helpless, keeping only practical interface needs. Checked that no persona's goals or abilities depend on culture or disability. The final personas are the team's own synthesis of both drafts, not a copy of either.

#	Tool	Exact prompt	What it gave	What we changed and why
3	Claude	Review these three personas for stereotypes in names, jobs, gender, culture, disability, and examples. List problems and suggested fixes in a table. Do not rewrite the personas.	A bias-check table with seven suggestions (run on 26 October 2026; full output in the table below).	Accepted four (1-4), rejected three (5-7). Accepted example: reframe Nour's difficulty as second-language reading and limited data, not culture. Rejected example: make Hana a novice AI user, because it removes the advanced persona that the coverage requirement needs. The other rejected suggestions changed goals without a reason.
4	Claude	I plan to use the OpenAlex public API for abstracts in a student project. What should I check about the license and data use before I download?	A checklist of license, terms of use, attribution, and rate limits.	Checked the official OpenAlex pages ourselves on 26 October 2026 and recorded the result in Section 3 (CC0 license, free API key, citation request). Did not rely on the AI answer alone.

Output of prompt 3 (bias review of the three draft personas)

#	Area	Problem found	Suggested fix	Decision
1	Culture and language	Nour's slow English reading and "first in her family" could read as a deficit linked to being Egyptian.	State the difficulty as second-language reading and limited mobile data, not culture.	Accepted
2	Disability	Hana's low vision could be read as a limitation. Her needs are listed as a problem to solve.	Describe the needs as interface requirements (zoom, screen reader, landmarks) and keep her an advanced, capable user.	Accepted
3	Connection and device	Limited data or poor Wi-Fi could be assumed from where a person lives.	Tie each limit to a situation (shared phone, Wi-Fi drops at peak hours), not to a place.	Accepted
4	Examples and tone	Topics and greetings could lean toward one cultural or regional context.	Use topics from pharmacy, engineering, and data science; use neutral greetings and a user-adjustable tone.	Accepted
5	Goals (Oliver)	His goal of a fast reading list may seem shallow.	Change his goal to a deep critical review of each paper.	Rejected: changes his goal without reason; speed under a deadline is his real need.

#	Area	Problem found	Suggested fix	Decision
6	AI experience (Hana)	Hana is the only experienced AI user and could be read as the "ideal" user, with novices as the problem.	Make Hana a novice AI user.	Rejected: removes the advanced persona the coverage requirement needs.
7	Goals (Nour)	Her goal is limited to understanding five papers.	Add a goal of publishing her own paper.	Rejected: unrealistic for a first-time student and changes her goal without reason.